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The Effect of The Blade Profile on Thrust of Electric Ducted Fan

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Abstract. An efficient and environmentally friendly mode of air transportation need electric engine. Electric Ducted Fan (EDF) is an engine thruster consisting of a duct, fan and electric motor which has a motor power source that comes from a battery so that it can produce compressed air which can then produce thrust. In this study, a variation of the blade profile was carried out for Electric Ducted Fan (EDF) with the aim of knowing the effect of the shape of the blade profile on the thrust that can be generated by the EDF. The profiles used are FX 74-CL5-140, Naca 64 (3)-418, and Naca 747A315 obtained from Airfoil Tools. The research method is to design and simulate the EDF with a variety of blade profiles that have been determined using the SOLIDWORKS 2021 software. The highest thrust results were obtained by the EDF with the Naca 747A315 blade profile variation, which is 31.5 Newton at 60000 RPM. The massflow value is carried out by numerical calculation, namely 0.7 kg/s. The higher the RPM value, the higher the value of thrust, velocity, and pressure obtained by the Electric Ducted Fan (EDF).

INTRODUCTION

Electric Ducted Fan (EDF) is an electric drive consisting of an electric motor connected to a fan that is integrated with the ducted tube [1]. The use of Electric Ducted Fan (EDF) has begun to be developed and applied to small-scale passenger aircraft. However, the current use or utilization of Electric Ducted Fan (EDF) is mostly applied to unmanned aircraft or UAV (Unmanned Aerial Vehicle) [2,3]. Unmanned Aerial Vehicle (UAV) or Unmanned Aircraft System (UAS) is an aircraft that can carry out missions independently using several sensors [4]. Besides being used in Unmanned Aerial Vehicle (UAV) or Unmanned Aircraft System (UAS), the use of Electric Ducted Fan (EDF) is also used in electric rockets [5,6]. Electric Ducted Fan (EDF) itself is composed of several components put together [7]. One of the important things to learn from the Electric Ducted Fan (EDF) system is the propulsion system or thrust [8]. Thrust is the thrust resulting from the propulsion of the aircraft engine by means of backward air so that the aircraft can go forward. In this study, variations of the blade profile [9] were carried out on the EDF, which used the FX 74-CL5-140, Naca 64(3)-418, and Naca 747A315 profiles obtained from the official Airfoil Tools website. The airfoil shape numbering system for the 64(3)-418 airfoil the last two digits indicate the airfoil thickness (18%). The first digit (6) is the airfoil series designation and the second digit indicates the chordwise position of minimum low pressure in tenths of chord (or the intention to have about 40% laminar flow). As long as the boundary layer stays laminar in the

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front of the airfoil, its drag is low and the range of this bucket in terms of ΔC_l is ±0.3 near the designed C_l of 0.4 (hence the subscript 3, and the digit 4 after the dash) [10].

The purpose of this study was to determine the effect of the blade profile on the thrust that can be generated by EDF. The research phase begins with determining the EDF reference parameters used. The EDF design process was carried out with various blades and carried out simulations with SOLIDWORKS 2021 software. Several literature reviews of fan profiles and electric propulsion systems were found in the following literature, for example, [11], regarding the design of blade profiles for gas turbine jet engines. [12], regarding EDF testing using a static thrust test tool with an experimental method. [7], regarding EDF motor thrust testing using test surface through negative pressure generation. This literature can provide an overview and understanding of profile selection and thrust analysis.

The purpose of this article is to simulate an Electric Ducted Fan using the SOLIDWORKS 2021 software, namely a flow simulation to find out the value of the thrust, speed and pressure that an Electric Ducted Fan can produce. To this goal, a novel computational setup for the EDF testing is proposed for acquiring important properties such as thrust force, velocity and generated pressure during the EDF's operation with different rotational velocity. In addition, this article will also contribute towards the novel evaluation of different design variables and modifications to the original EDF blade or fan structure for their effect on the thrust productivity and the dependence of the airfoil design to the EDF's blade performance.

Articles are arranged in the following order: 1. Introduction and research background, 2. Data collection methodology, 3. Results and discussion of thrust, speed, and pressure, 4. Conclusion, notation list, and bibliography.

METHODS

In this research, a design and simulation process was carried out to find out several parameter values of the thrust, speed, and pressure that occur in the EDF. The 90 mm FMS Electric Ducted Fan (EDF) for model simulation is shown in Figure 1.

FIGURE 1. EDF FMS 90 mm

The reference object used in this study is an Electric Ducted Fan (EDF) with a sheath diameter of 90 mm, a blade diameter of 89.5 mm, and a total of 12 blades. For the specifications of the 3546 1950 kV series electric motor.

Before the simulation process is carried out, the blade design stage is carried out first. The various blade profiles used are FX 74-CL5-140, Naca 64(3)-418, and Naca 747A315. The design of the blade with various profiles is shown in Figure 2. Blade FX 74 CL5 140, Figure 3. Blade Naca 64(3)- 418, Figure 4. Blade Naca 747A315.

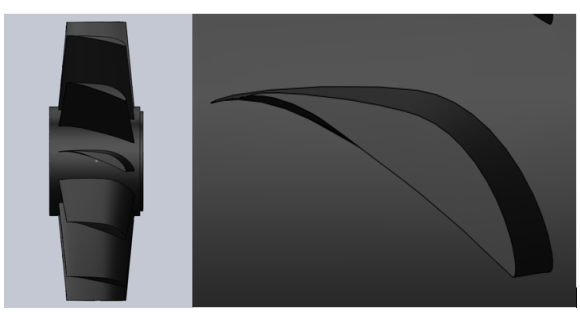

FIGURE 2. FX 74 CL5 140 blades

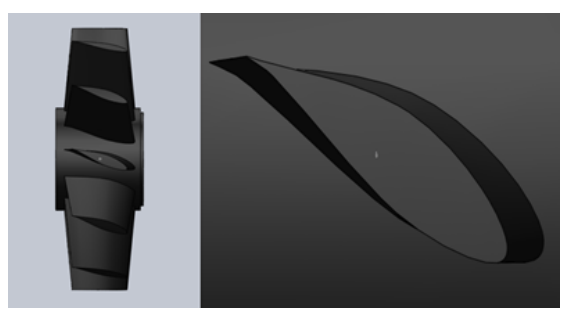

FIGURE 3. Naca Blade 64(3)-418

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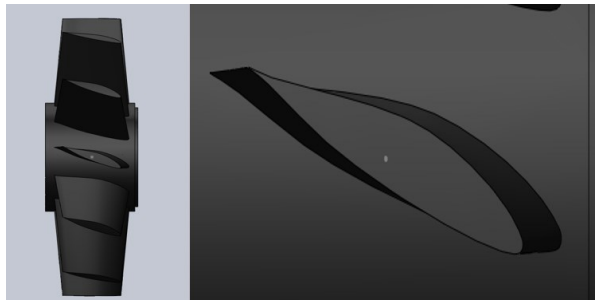

FIGURE 4. Naca Blade 747A315

The NACA 7-series wing sections are characterized by a greater extent of possible laminar flow on the lower than on the upper surface. These sections permit low pitching-moment coefficients with moderately high design lift coefficients at the expense of some reduction of maximum lift and critical Mach number.

The NACA 747A315 in Fig.4 explain the first number 7 indicates the series number. The second number 4 indicates the extent over the upper surface, in tenths of the chord from the leading edge, of the region of favorable pressure gradient at the design lift coefficient. The third number 7 indicates the enent over the lower surface, in tenths of the chord from the leading edge, of the region of favorable pressure gradient at the design lift coefficient. The significance of the last group of three numbers is the same as for the NACA 6-series wing sections. The letter A which follows the first three numbers is a serial letter to distinguish different sections [13].

After the blade design process is complete, the next step is the blade assembly process with other EDF components so that it becomes a complete EDF. Then a simulation process is carried out with the SOLIDWORKS 2021 software, namely external flow simulation. This iterative process will generate a pack of data about flow, pressure, and thrust. The simulation result of flow velocity inlet on the EDF can be seen in Figure 5.

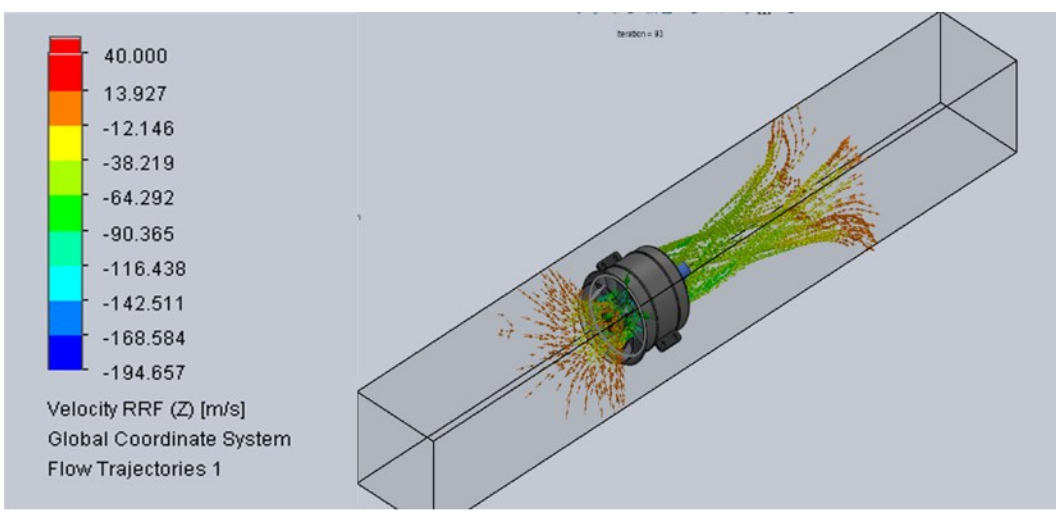

FIGURE 5. Flow simulation result for Naca 64(3)-418 EDF in the SOLIDWORKS 2021

RESULTS AND DISCUSSION

After the simulation process is complete and the result data is obtained, the next step is to discuss the results of the simulation that has been obtained.

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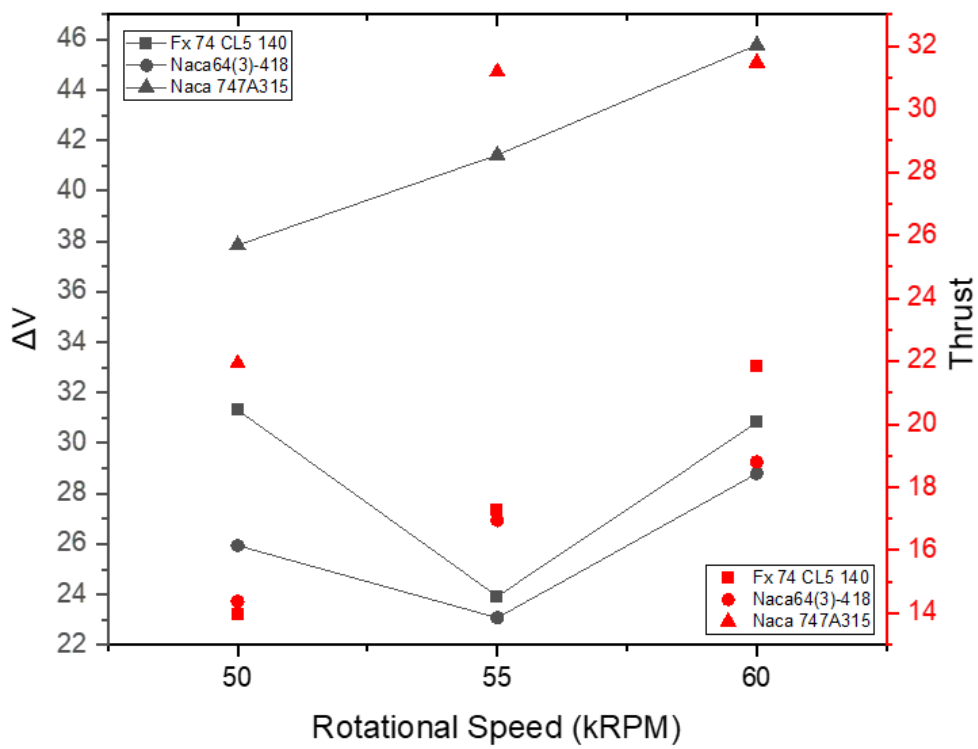

FIGURE 6. Graph of the relationship between delta velocity and thrust towards angular speed of ducted fan

In Figure 6 it can be seen on the EDF with the FX 74-CL5-140 blade profile at 50000 RPM it gets a thrust value of 13.94 N, inlet and outlet flow velocity of 26.35 m/s and 57.66 m/s, so it produces delta flow velocity around 31 point. At angular speed of 55 kRPM it produces a thrust value of 17.27 N, inlet and outlet flow velocity of 35.02 m/s and 58.93 m/s, the *delta* flow velocity around 24 point. Then, at ω of 60 kRPM it produces a thrust value of 21.84 N and increment of flow velocity ΔV about 30.82 m/s.

For EDF with a blade profile Naca 64(3)-418 at 50000 RPM it produces a thrust value of 14.37 N, inlet and outlet flow velocity of 28.62 m/s and 54.55 m/s, so it produces delta flow velocity around 26 point. At angular speed of 55 kRPM it produces a thrust value of 16.95 N, inlet and outlet flow velocity of 39.95 m/s and 57.02 m/s, the *delta* flow velocity around 23 point. Then, at ω of 60 kRPM it produces a thrust value of 18.80 N and increment of flow velocity ΔV about 28.79 m/s.

Meanwhile for EDF with the airfoil type of Naca 747A315 at 50000 RPM it gets a thrust value of 21.92 N, inlet and outlet flow velocity of 45.25 m/s and 83.10 m/s, so it produces delta flow velocity around 38 point. At angular speed of 55 kRPM it produces a thrust value of 31.18 N, inlet and outlet flow velocity of 50.78 m/s and 92.20 m/s, the *delta* flow velocity around 41 point. Then, at ω of 60 kRPM it produces a thrust value of 31.47 N and increment of flow velocity ΔV about 45.79 m/s.

From the graph of the relationship between flow velocity and rotational or angular speed in Figure 6, it can be concluded that the higher of the ω (*omega*) value, the higher the flow velocity value that can be generated by the EDF.

Similarly, the graph of the relationship between thrust and angular speed of ducted-fan in Figure 6, it can be culminated that the higher the *omega* value, the higher the thrust of propulsion that can be achieved by the Electric Ducted Fan.

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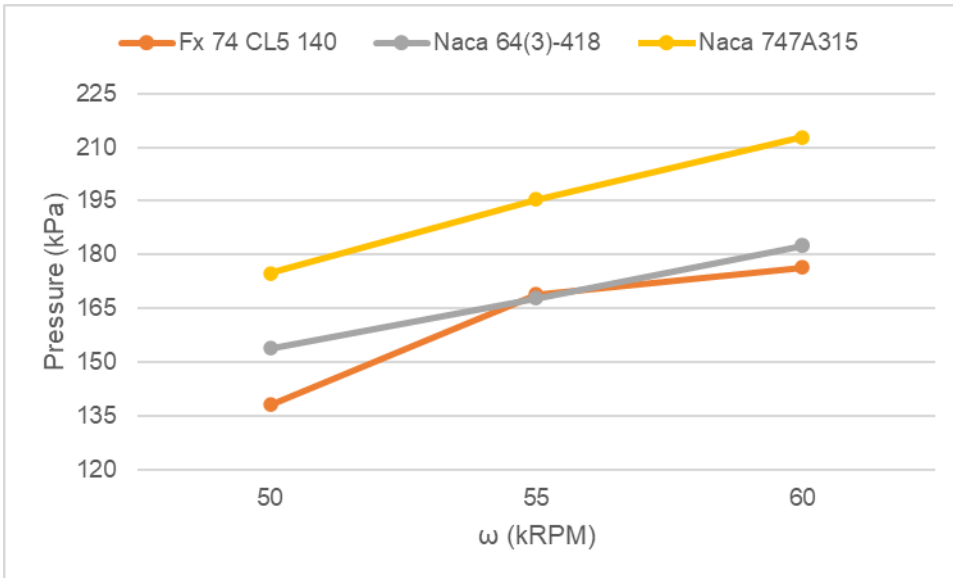

FIGURE 7. Graph of the relationship between pressure and RPM

In Figure 7 it can be seen that the EDF with the blade profile FX 74-CL5-140 at 50000 RPM gets a pressure value of 138173 Pa, at 55000 RPM it gets a pressure value of 168888 Pa, and at 60000 RPM it gets a pressure value of 176442 Pa. For EDF with a blade profile Naca 64(3)-418 at 50000 RPM it gets a pressure of 153820 Pa, at 55000 RPM it gets a pressure value of 167914 Pa, and at 60000 RPM it gets a pressure of 182593 Pa. Whereas for EDF with a Naca 747A315 blade profile at 50000 RPM it gets a pressure value of 174777 Pa, at 55000 RPM it gets a pressure value of pressure of 195456 Pa, and at 60000 RPM get a pressure of 212855 Pa.

These results lead to discussion about how much thrust can be produced by EDF with different methods. From [12] experimental setup from Ø 80 mm of EDF give thrust around 31 N. Different experimental setup has been done by [7] for Ø 92 mm Dr. MadThrust Co EDF provides a maximum thrust of 37 N and 35 – 55 N adhesion force. 3D printed EDF [14] has been built by van Rooij to perform hi speed aircraft. This research gives insight wind tunnel procedure for thrust production of NACA 65-010 airfoil EDF. The theoretical produced thrust at 208.3 m/s and 42 kRPM is 95.1 N with a required power input of 27.8 kW. For the low-speed prototype this is 15.1 N at a speed of 84 m/s, 17 kRPM and a rotor input power of 1.97 kW. And this research of computational method for EDF thrust system with different rotational speeds and airfoil blades lead to the results variant from 19, 22, and 31.5 N.

From the graph of the relationship between pressure and rpm in Figure 9, it can be concluded that the higher the RPM value, the higher the pressure obtained by the EDF.

To find out the value of massflow (kg/s), take one of the data obtained from the simulation process, namely data from the Naca 747A315 profile simulation at 60000 RPM. The calculation uses the following equation.

Is known:

$$T = 31.475\ N$$
$$V_o = 97.65\ m/s$$
$$V_i = 51.86\ m/s$$

$$\begin{aligned} T &= \dot{m}(V_o - V_i) \\ 31.475 &= \dot{m}(97.65 - 51.86) \\ \dot{m} &= 0.68\ kg/s \end{aligned} \tag{1}$$

The massflow value is carried out by numerical calculation, namely 0.7 kg/s.

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CONCLUSION

After obtaining data from the simulation process and analysis has been carried out so as to obtain such results, the next step is to draw conclusions. From the results above it can be concluded that:

The highest thrust and velocity values at 50000 RPM, 55000 RPM and 60000 RPM were generated by an Electric Ducted Fan with a Naca 747A315 blade profile, with thrust values of 22 N, 31.2 N and 31.5 N respectively. The outlet flow velocity values respectively 83.1 m/s, 92.2 m/s and 97.7 m/s.

The highest pressure was obtained by Electric Ducted Fan with the Naca 747A315 blade profile around 176 - 213 kPa respectively.

The massflow value is carried out by numerical calculation, namely 0.7 kg/s. The higher the RPM value, the higher the value of thrust, velocity, and pressure obtained by the Electric Ducted Fan (EDF).

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